

QuitTxt RAG Evaluation

Technical Implementation Brief — March 2026

This document provides a comprehensive technical summary of the RAG (Retrieval-Augmented Generation) evaluation experiments conducted for the QuitTxt smoking cessation AI counselor project. It covers experimental design, datasets, methods, all results, and conclusions — intended as a briefing document for team meetings and academic publication preparation.

Project	QuitTxt Protocol — AI Smoking Cessation Counselor
Research Question	Does dataset curation quality affect RAG faithfulness?
Model	GPT-4o-mini (gpt-4o-mini)
Evaluation Framework	RAGAS 0.4.x (collections API)
Test Set	127 questions (LFV-reviewed, from Dr. Louis Willis's 150-question set)
Date Range	February 19 – March 11, 2026

1. Background & Research Context

The QuitTxt project builds an AI-powered smoking cessation counselor that uses Retrieval-Augmented Generation (RAG) to ground responses in curated knowledge. Three knowledge sources were compared: AI-generated QA pairs, human-curated QA pairs, and web-scraped content. The central research question is whether the **quality and origin of the knowledge base** affects the faithfulness and reliability of the AI's responses.

A secondary problem emerged during analysis: the evaluation model (GPT-4o-mini) was generating responses 3–4× longer than the retrieved context, extrapolating beyond what the knowledge base contained. This verbosity problem suppressed faithfulness scores and was addressed in a second experimental phase.

1.1 Reference Benchmark (Ebrahim et al.)

The study references the Ebrahim et al. baseline: Base LLM faithfulness = 0.58 (n=10), WebRAG faithfulness = 0.86 (n=100). Our experiments aim to contextualise and extend these findings with human-curated and AI-generated datasets.

2. Datasets

Dataset	Size	Source	Format	Location
AI-Generated QA	5,936 pairs	Gemini 2.0 Flash (faithfulness-constrained)	JSON	data/ai_generated/ai_generated_qa_full.json
Human-Curated QA	4,431 pairs	Manual curation	JSONL	qa.jsonl (project root)
Web-Scraped QA	278 pairs	Web scraping pipeline	JSONL	data/web_scraped_qa.jsonl
Test Set	127 questions(150Q, 23 excluded by reviewer)	Dr. Louis Willis	XLSX	data/test_set/150_qa_testset.xlsx

The test set uses 127 of Dr. Willis's 150 questions — 23 were flagged by reviewer LFV (column 83 = Comments LFV) and excluded to ensure clean ground-truth references. Each question has a ground_truth answer used by RAGAS for context precision and recall scoring.

2.1 Dataset Answer Length (stored in ChromaDB)

The chromaDB knowledge base documents were constructed as 'Q: [question]\nA: [answer]' strings. Analysis of the stored answers showed the following average lengths:

Dataset	Avg Answer Words	Observation
AI-Generated	~32 words	Short, faithful to source excerpts
Human-Curated	~39 words	Slightly longer, practitioner language
Web-Scraped	~54 words	Longer passages, noisier content

3. System Architecture & Methods

3.1 RAG Pipeline

Each evaluation configuration follows the same pipeline:

- **Retrieval:** ChromaDB (ephemeral, cosine similarity, HNSW index) retrieves top-3 documents for each test question
- **Generation:** GPT-4o-mini generates an answer using the retrieved contexts as a knowledge base
- **Evaluation:** RAGAS 0.4.x scores the (question, answer, context, reference) tuple on 4 metrics
- **Checkpointing:** Each answer is appended to a JSONL checkpoint immediately, enabling resume on failure

3.2 Evaluation Configurations

Config	Knowledge Source	Context Used?	Notes
baseline	None (LLM only)	No	Pure GPT-4o-mini, no retrieval
ai_rag	AI-Generated QA (5,936)	Yes	Gemini-generated, faithful to source
human_rag	Human-Curated QA (4,431)	Yes	Manually curated QA pairs
web_rag	Web-Scraped QA (278)	Yes	Small dataset, noisy content

3.3 RAGAS Metrics

Metric	Measures	Input Required	Range
Faithfulness	Are all claims in the response supported by the context?	question, response, contexts	0–1 (higher = better)
Answer Relevancy	Is the response relevant to the question?	question, response	0–1
Context Precision	Are retrieved chunks relevant to the reference answer?	question, reference, contexts	0–1
Context Recall	Does the retrieved context cover the reference answer?	question, reference, contexts	0–1

Key metric: Faithfulness is the primary metric throughout this study because it directly measures hallucination risk — whether the AI makes claims not supported by what it retrieved. Baseline (no context) faithfulness is always NaN because there is no context to evaluate against.

3.4 Technical Implementation

- **Model:** GPT-4o-mini — temperature=0.3, max_tokens=300 (verbose) / 150 (concise)
- **Embedding:** ChromaDB default (all-MiniLM) for retrieval; OpenAI embeddings for AnswerRelevancy scoring

- **RAGAS LLM:** GPT-4o-mini via AsyncOpenAI, max_tokens=8192 (required for verbose Faithfulness JSON output)
- **Rate limiting:** 65s cooldown between metrics, 120s between configs (200K TPM limit on gpt-4o-mini)
- **Script:** scripts/ragas_evaluation.py — supports --configs, --max-questions, --resume, --concise flags

4. The Verbosity Problem (Key Finding)

A critical issue was identified during early evaluation: GPT-4o-mini was generating responses **3–4× longer** than the retrieved context. When the model has only ~32–54 words of context but generates 127–139 word responses, it must *invent* content — producing unsupported claims that RAGAS correctly penalises as hallucinations.

	AI-RAG	Human-RAG	Web-RAG
Context avg words (stored)	~32 words	~39 words	~54 words
Verbose response avg words	127.6 words	133.1 words	139.2 words
Ratio (response / context)	4.0×	3.4×	2.6×
Concise response avg words	30.6 words	39.2 words	34.0 words
Reduction from concise	–76%	–71%	–76%

4.1 The Fix: --concise Mode

Two prompt changes were implemented under the --concise flag:

- **Verbose prompt:** 'Base your answer on the knowledge provided.' — permissive, allows supplementing with training data
- **Concise RAG prompt:** 'Answer in 2-3 sentences using ONLY facts from the knowledge base below. Do NOT add information, examples, or details not explicitly stated in the knowledge base.'
- **max_tokens reduced:** 300 → 150 (hard cap prevents verbose output even if model tries)

Concise checkpoints are stored separately (checkpoint_concise_*) so verbose and concise results can be directly compared without overwriting previous data.

5. Experiments Conducted

#	Date	Mode	n	Configs	Purpose
1	Feb 19	Verbose	20	baseline, ai_rag, human_rag	First n=20 benchmark
2	Feb 25	Verbose	5	ai_rag	Debugging sanity check
3	Feb 26	Verbose	100	baseline, ai_rag, human_rag	Full verbose baseline
4	Feb 26	Verbose	100	human_rag only	Human-RAG standalone
5	Mar 2	Verbose	5/20/100	web_rag	Add web-scraped config
6	Mar 9	Concise	5	ai_rag	Smoke test: verify word reduction
7	Mar 9	Concise	20	baseline, ai_rag, web_rag	Concise trial (3 configs)
8	Mar 10	Concise	20	human_rag	Complete concise comparison
9	Mar 10	Concise	100	ai_rag, human_rag, web_rag	Full n=100 concise run

6. Results

6.1 Verbose Mode — Faithfulness (Primary Metric)

Config	n	Mean $\pm$ SD	Hallucination Rate	vs Ebrahim Baseline
Baseline (no RAG)	—	NaN (expected)	—	Ebrahim Base: 0.58
AI-RAG	20	0.564 $\pm$ 0.328	43.6%	−0.016 vs Base
AI-RAG	100	0.531 $\pm$ 0.320	46.9%	−0.049 vs Base
Human-RAG	20	0.755 $\pm$ 0.298	24.5%	+0.175 vs Base
Human-RAG	100	0.657 $\pm$ 0.334	34.3%	+0.077 vs Base
Web-RAG	20	0.528 $\pm$ 0.304	47.2%	−0.052 vs Base
Web-RAG	100	0.514 $\pm$ 0.350	48.6%	−0.066 vs Base

Interpretation (Verbose): Human-RAG outperforms AI-RAG and Web-RAG in verbose mode, but all configs are suppressed by verbosity-induced hallucination. AI-RAG and Web-RAG barely surpass the Ebrahim baseline — suggesting the verbose prompt negated the retrieval benefit.

6.2 Concise Mode — Faithfulness

Config	n	Mean $\pm$ SD	Median	Halluc. Rate	High Risk%
AI-RAG	5 (smoke)	0.920 $\pm$ 0.179	1.000	8.0%	0%

AI-RAG	20	0.777 ± 0.405	1.000	22.3%	20%
AI-RAG	100	0.724 ± 0.376	1.000	27.6%	20%
Human-RAG	20	0.946 ± 0.139	1.000	5.4%	0%
Human-RAG	100	0.789 ± 0.352	1.000	21.1%	17%
Web-RAG	20	0.799 ± 0.365	1.000	20.1%	15%
Web-RAG	100	0.719 ± 0.376	1.000	28.1%	19%

★ Green row = primary publishable result

6.3 Head-to-Head Comparison (Verbose vs Concise, n=100)

Config	Verbose n=100	Concise n=100	Absolute Gain	Relative Gain
AI-RAG	0.531	0.724	+0.193	+36.3%
Human-RAG	0.657	0.789	+0.132	+20.1%
Web-RAG	0.514	0.719	+0.205	+39.9%

7. Interpretation of Results

7.1 What Each Result Means

Verbose AI-RAG (0.531, n=100)

Near-baseline performance. Despite having 5,936 QA pairs to retrieve from, the model generates so much additional text that RAGAS detects ~47% of responses as containing unsupported claims. The RAG system is barely helping.

Verbose Human-RAG (0.657, n=100)

Better than AI-RAG, but still suppressed. Human context quality is higher, giving the model better material to stay grounded — but verbosity still causes ~34% hallucination rate.

Verbose Web-RAG (0.514, n=100)

The weakest performer. Small dataset (278 pairs) means poorer retrieval coverage, and the model fills gaps with fabricated content.

Concise AI-RAG (0.724, n=100)

A +36% improvement from verbose. Strict prompting and token cap eliminate much of the extrapolation. This is a viable, deployable configuration.

Concise Human-RAG (0.789, n=100) ★

The best-performing configuration overall. Human-curated content + concise prompting achieves 0.789 faithfulness — approaching Ebrahim's WebRAG (0.86) benchmark. Hallucination rate drops to 21%, with the tightest SD when measured at n=20 (SD=0.139).

Concise Web-RAG (0.719, n=100)

Improved from verbose but remains limited by small dataset size. Low retrieval coverage means the model still encounters questions it can't answer from context.

n=20 vs n=100 Divergence

n=20 scores (especially human_rag at 0.946) are optimistically biased — small samples happen to draw 'easy' questions. n=100 is the statistically reliable figure. Always cite n=100 in publications.

7.2 The Hierarchy

At n=100 concise mode (the most reliable results):

Rank	Config	Faithfulness	Interpretation
1st ★	Human-RAG (concise)	0.789	Best: high-quality context + strict prompt
2nd	AI-RAG (concise)	0.724	Good: large dataset compensates for lower curation
3rd	Web-RAG (concise)	0.719	Limited by small dataset size (278 pairs)

—	Ebrahim WebRAG ref.	0.860	External benchmark (different test set)
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8. Conclusions

1. Dataset curation quality matters — but prompting matters more

In verbose mode, human-curated RAG (0.657) outperforms AI-generated RAG (0.531) by 0.126. But concise prompting improves all configs by 0.13–0.21. The biggest lever is not which dataset you use, but how strictly you constrain the generation.

2. The verbosity problem is real and fixable

Responses 3–4× longer than context are a systematic flaw in permissive RAG prompts. The fix — explicit sentence limits + 'ONLY facts from the knowledge base' instruction + max_tokens cap — is simple, free, and highly effective.

3. Human-RAG is the best configuration for production deployment

At n=100 concise, human-RAG achieves 0.789 faithfulness with 21% hallucination rate. For a health application like QuitTxt, where false medical claims carry real risk, this is the recommended production configuration.

4. AI-Generated RAG is a viable fallback

At 0.724 concise faithfulness, AI-RAG is a practical option — especially given its larger dataset size (5,936 pairs) providing better retrieval coverage. The 0.065 gap versus human-RAG may be acceptable in contexts where human curation is costly.

5. n=20 results are unreliable; use n=100

Human-RAG showed 0.946 at n=20 — an optimistic outlier. The n=100 value of 0.789 is the correct figure for publication. Always validate small-sample results before reporting.

6. Context precision scores are incomplete

Rate limiting (200K TPM) caused context_precision failures for human_rag and web_rag at n=100. These need re-running with 120s inter-metric cooldowns instead of 65s.

9. Recommended Next Steps

Priority	Action	Command / Notes
HIGH	Fix context_precision rate limit (increase cooldown to 120s)	Increase _METRIC_COOLDOWN from 65 to 120 in ragas_evaluation.py, re-run n=100 concise
HIGH	Run full n=127 concise (all questions)	python3 scripts/ragas_evaluation.py --concise --configs ai_rag human_rag web_rag
MEDIUM	Add confidence intervals to results	Bootstrap CI or t-test for Human vs AI gap
MEDIUM	Add web_rag dataset expansion	More web-scraped pairs → better retrieval coverage
LOW	Run baseline concise (n=100)	For completeness in the paper table

10. Key Files & Commands Reference

10.1 Script Commands

Sanity check (n=5, ai_rag, concise):

```
python3 scripts/ragas_evaluation.py --max-questions 5 --concise --configs ai_rag
```

n=20 trial (all 3 RAG configs, concise):

```
python3 scripts/ragas_evaluation.py --max-questions 20 --concise --configs ai_rag
human_rag web_rag
```

Full n=100 run (concise):

```
python3 scripts/ragas_evaluation.py --max-questions 100 --concise --configs ai_rag
human_rag web_rag
```

Full n=127 (all questions, concise):

```
python3 scripts/ragas_evaluation.py --concise --configs ai_rag human_rag web_rag
```

Resume interrupted run:

```
python3 scripts/ragas_evaluation.py --concise --configs ai_rag --resume
```

10.2 Output Files

File Pattern	Contents
results/ragas_evaluation/ragas_eval_YYYYMMDD.json	Verbose run results (JSON)
results/ragas_evaluation/ragas_eval_concise_YYYYMMDD.json	Concise run results (JSON)
results/ragas_evaluation/checkpoint_{config}_{n}.jsonl	Verbose answers per question
results/ragas_evaluation/checkpoint_concise_{config}_{n}.jsonl	Concise answers per question
results/TECHNICAL_BRIEF_Mar2026.pdf	This document